

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Uniform indentation maintained across all Python modules
2	Proper File Structure	Files and folders are logically organized	yes	Project organized into data, models, dashboard, and utility modules
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Variables clearly describe emotions, predictions, and recommendations
4	Function / Method Names	Functions are descriptively named	yes	Functions named according to their purpose
5	Code Comments	Inline and block comments explain logic	yes	Important logic and processing steps are documented
6	Modular Design	Code is split into reusable functions/modules	yes	Emotion detection, recommendation generation, and dashboard are separated into modules
7	No Redundant Code	Duplicate or unused code is removed	yes	Reusable functions reduce duplication
8	Error Handling	Exceptions and errors are handled gracefully	yes	Input validation and exception handling implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Emotion Detection Module	Python, BiLSTM, BERT	Emotion prediction workflow	high

2	Recommendation Engine	Gemini API, Python	Personalized learning guidance	high
3	Data Preprocessing Module	Python, NLTK	Text cleaning and preparation	high
4	Streamlit Dashboard	Streamlit	Student interaction and analytics	high
5	Emotion History Storage	CSV, Python	Progress tracking and reporting	high

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized architecture with separate modules
Readability	5	Clear naming conventions and documentation
Reusability	5	Reusable AI, NLP, and dashboard components
Documentation / Comments	4	Sufficient comments and project documentation
Overall Score	4.7	Maintainable, scalable, and easy to extend